

Heaven's Light Is Our Guide



Rajshahi University of Engineering & Technology **Department of Electrical & Computer Engineering**

Course Title

Digital Techniques Sessional

Course No: ECE 2112

Lab No: 01

Report Title:

Implementation of Logic Gates Using Universal Gates, Full Adder, and Binary to BCD Converter

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Problem: 01

Problem Definition: Implementation of an OR gate using only NOR gates.

Objective:

To design and verify an OR gate using two NOR gates, and to observe how universal gates can replicate the functionality of other basic gates.

Truth Table:

A	B	G1	$Y = A + B$
0	0	1	0
0	1	0	1
1	0	0	1
1	1	0	1

Table 1: Caption

Circuit Diagram:

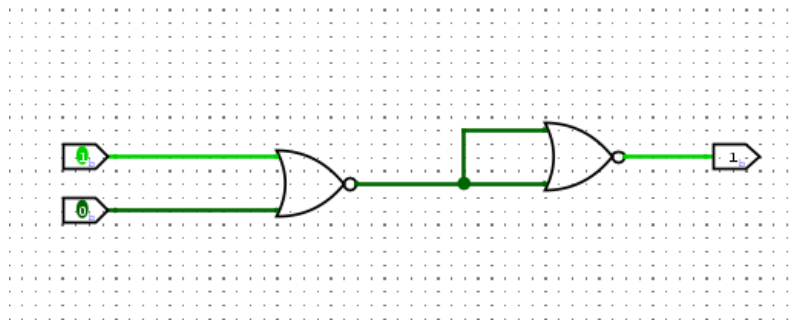


Figure 1: OR using NOR

Problem: 02

Problem Definition: Implementation of an OR gate using only NAND gates.

Objective:

To design and verify an OR gate using three NAND gates, applying De Morgan's theorem.

Truth Table:

A	B	$\bar{A}$	$\bar{B}$	$Y = A + B$
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	1

Table 2: Caption

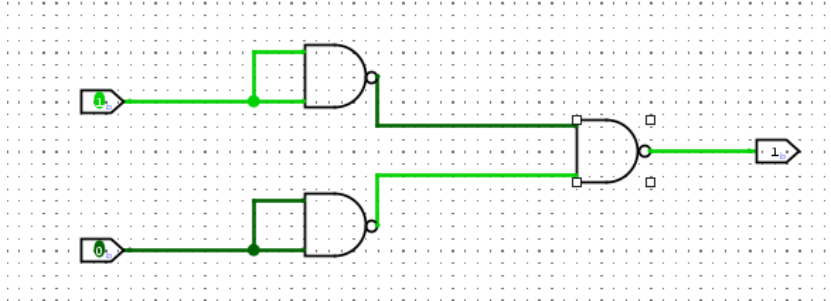
Circuit Diagram:

Figure 2: OR using NAND

Problem: 03

Problem Definition: Implementation of an AND gate using only NOR gates.

Objective: To design and verify an AND gate using three NOR gates, applying De Morgan's theorem.

Truth Table:

A	B	$\bar{A}$	$\bar{B}$	$Y = A \cdot B$
0	0	1	1	0
0	1	1	0	0
1	0	0	1	0
1	1	0	0	1

Table 3: Caption

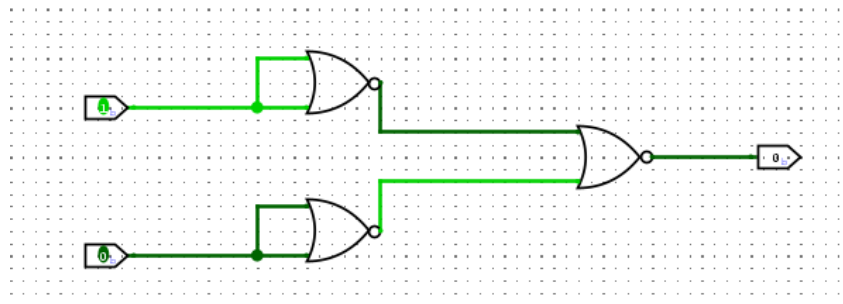
Circuit Diagram:

Figure 3: AND using NOR

Problem: 04

Problem Definition: Implementation of an AND gate using only NAND gates.

Objective:

To design and verify an AND gate using two NAND gates.

Truth Table:

A	B	$G1 = \bar{A} \cdot \bar{B}$	$Y = A \cdot B$
0	0	1	0
0	1	1	0
1	0	1	0
1	1	0	1

Table 4: Caption

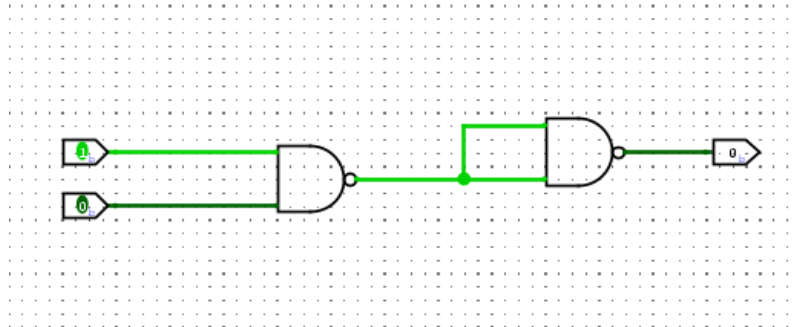
Circuit Diagram:

Figure 4: AND using NAND

Problem: 05

Problem Definition: Implementation of a NOT gate using only a NOR gate.

Objective:

To design and verify a NOT gate (inverter) using a single NOR gate.

Truth Table:

A	$Y = \bar{A}$
0	1
1	0

Table 5: Caption

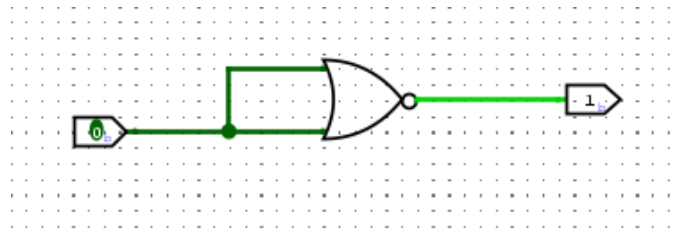
Circuit Diagram:

Figure 5: NOT using NOR

Problem: 06

Problem Definition: Implementation of a NOT gate using only a NAND gate.

Objective:

To design and verify a NOT gate (inverter) using a single NAND gate.

Truth Table:

A	$Y = \bar{A}$
0	1
1	0

Table 6: Caption

Circuit Diagram:

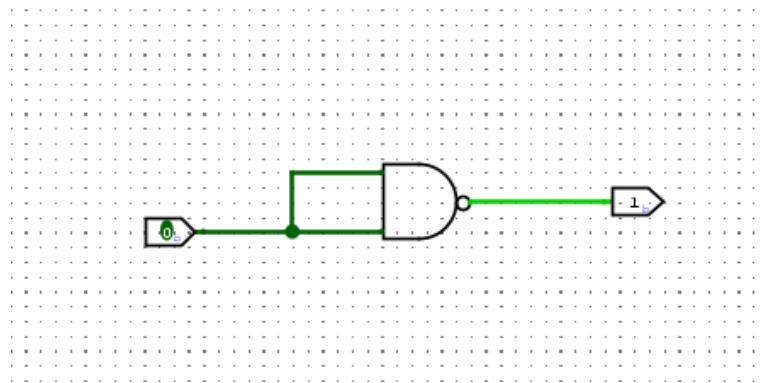


Figure 6: NOT using NAND

Problem: 07

Problem Definition: Implementation of a full adder circuit and testing it against its truth table.

Objective:

To design a full adder using basic logic gates and verify its Sum and Carry outputs for all eight possible input combinations of A, B, and Cin.

Truth Table:

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Table 7: Caption

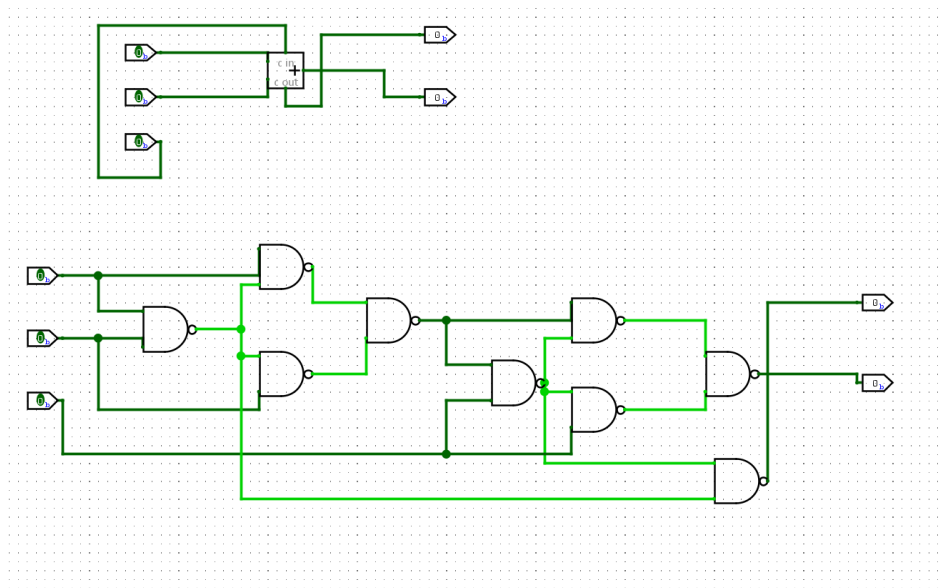
Circuit Diagram:

Figure 7: FULL ADDER Circuit

Problem: 08

Problem Definition: Implementation of a binary to BCD converter for a 4-bit binary input.

Objective:

To design a combinational logic circuit that converts a 4-bit binary number (representing decimal values 0 to 15) into its equivalent two-digit Binary Coded Decimal (BCD) representation.

Truth Table:

B3	B2	B1	B0	T	U3	U2	U1	U0
0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	1
0	0	1	0	0	0	0	1	0
0	0	1	1	0	0	0	1	1
0	1	0	0	0	0	1	0	0
0	1	0	1	0	0	1	0	1
0	1	1	0	0	0	1	1	0
0	1	1	1	0	0	1	1	1
1	0	0	0	0	1	0	0	0
1	0	0	1	0	1	0	0	1
1	0	1	0	1	0	0	0	0
1	0	1	1	1	0	0	0	1
1	1	0	0	1	0	0	1	0
1	1	0	1	1	0	0	1	1
1	1	1	0	1	0	1	0	0
1	1	1	1	1	0	1	0	1

Table 8: Caption

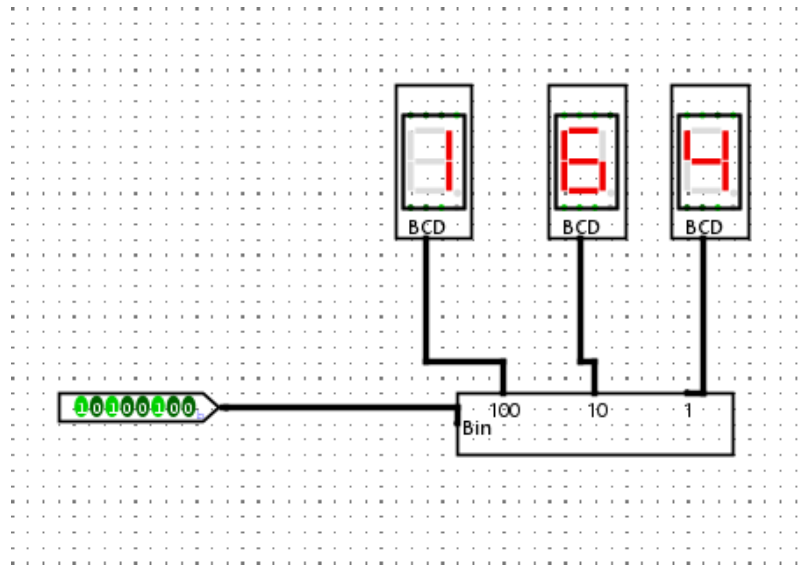
Circuit Diagram:

Figure 8: Binary to BCD

References

1. M. M. Mano, Digital Design, 5th ed. Upper Saddle River, N.J.: Pearson, 2013.
2. R. J. Tocci, N. S. Widmer, and G. L. Moss, Digital Systems: Principles and Applications, 11th ed. Upper Saddle River, N.J.: Pearson, 2011.